

Material and Methods

Theoretical Background

The Proximity framework assumes that rhythmic organization in animal signaling emerges from intentional or behaviorally regulated placement of acoustic events in time. If such regulation exists, the inter-onset intervals (IOIs) of a sequence are expected to cluster around discrete temporal targets rather than being randomly or mechanically determined. Each cluster can be approximated by a Gaussian distribution, reflecting natural motor and perceptual variability around a behavioral timing target.

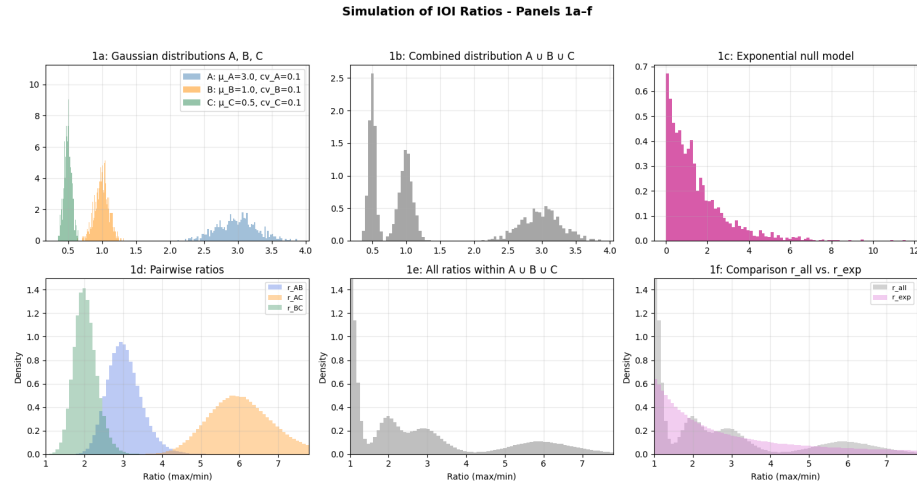


Figure 1: Illustration of ratio distributions from simulated Gaussian IOIs and a null exponential model. **Top row:** (1a) Gaussian distributions A, B, C (color-coded), (1b) combined distribution $A \cup B \cup C$, (1c) exponential distribution with the same mean. **Bottom row:** (1d) pairwise ratios r_{AB}, r_{AC}, r_{BC} , (1e) ratios from combined distribution r_{all} , (1f) comparison of combined ratios with exponential null model.

We denote three independent Gaussian distributions as A, B, C , with respective means μ_A, μ_B, μ_C and standard deviations $\sigma_A, \sigma_B, \sigma_C$. Let the values in each distribution be $a_i \in A$, $b_j \in B$, and $c_k \in C$.

For each pair of distributions, we compute the ratio for all possible pairs of values:

$$r_{AB} = \frac{\max(a_i, b_j)}{\min(a_i, b_j)}, \quad a_i \in A, b_j \in B,$$

$$r_{AC} = \frac{\max(a_i, c_k)}{\min(a_i, c_k)}, \quad a_i \in A, c_k \in C,$$

$$r_{BC} = \frac{\max(b_j, c_k)}{\min(b_j, c_k)}, \quad b_j \in B, c_k \in C.$$

This procedure ensures that all cross-cluster ratios are included. Fig. 1a illustrates the three Gaussian distributions, and Fig. 1d shows the resulting pairwise ratio distributions. These distributions are approximately Gaussian, with a slight left skew.

In empirical sequences, it is generally not possible to determine in advance to which cluster an IOI belongs. Therefore, we compute **all ratios within the combined distribution** $A \cup B \cup C$:

$$r_{\text{all}} = \frac{\max(x_i, x_j)}{\min(x_i, x_j)}, \quad x_i, x_j \in A \cup B \cup C.$$

The distribution r_{all} now includes all same-cluster ratios, resulting in an additional curve centering around 1. This new “peaks” spread reflects the variance of the underlying Gaussian distributions (Fig. 1b,e).

Null model concept

To provide a baseline for random, unstructured timing, we compare the ratio distribution of a sequence to that of a random (exponential) distribution with the same mean as the observed IOIs (Fig. 1c,f). This null model represents a scenario of maximal randomness, where inter-onset intervals are independently drawn from

$$E \sim \text{Exp}(\lambda), \quad \lambda = \frac{1}{\bar{x}},$$

with $\bar{x}$ being the mean of the empirical IOIs. Ratios are then computed analogously as

$$r_E = \frac{\max(e_i, e_j)}{\min(e_i, e_j)}, \quad e_i, e_j \in E.$$

Proximity Function and Parameters – Gaussian Kernels

After building these empirical IOI ratio distributions (and their respective null distributions), we score them on a theoretically expected structure, the Proximity function $P(x)$. This function scores how well observed IOI ratios align with simple integer ratios, approximating the ratio distributions of two Gaussian IOI clusters as Gaussian kernels centered at rational values k/f .

Formally, the Proximity function is defined as

$$P(x) = \max_{\substack{1 \leq f \leq f_{\text{max}} \\ k \geq 1}} \frac{1}{f^{w_e} \cdot (k/f)^d} \cdot \exp \left[-\frac{(x - k/f)^2}{2\sigma_{f,k}^2} \right],$$

with peak widths

$$\sigma_{f,k} = \frac{1}{S_b \cdot f^{s_e} \cdot k^{S_g}}.$$

Here, $P(x) \in [0, 1]$ quantifies the closeness of the observed ratio x to a simple integer ratio. The function consists of two key components:

- **Height term:** $\frac{1}{f^{w_e}(k/f)^d}$, which down-weights contributions from ratios with higher denominators f or numerators k , reflecting perceptual salience of simple integer ratios.
- **Width term:** $\sigma_{f,k}$, which defines the expected spread of each Gaussian peak, capturing the variability around the ideal ratio due to natural motor and perceptual noise.

Empirical IOI ratios that cluster around these Gaussian peaks will receive high proximity scores, indicating that the underlying IOI clusters likely form small-integer relationships (Fig. 1d,e). If IOI clusters are unrelated or randomly distributed, the ratios will many times not align with the peaks, resulting in low proximity values.

Parameters

- x (**IOI ratio**): Observed ratio of two inter-onset intervals within a sequence.
- f (**harmonic order**): Integer frequency multiplier. Each f corresponds to a possible subdivision or multiple of the underlying beat.
- k (**integer multiple**): Numerator of the ratio k/f . Each peak in the function corresponds to such a rational value.
- S_b (**sharpness_base**): Baseline sharpness of peaks. Higher S_b produces narrower peaks, rewarding ratios closer to exact integers.
- S_g (**sharpness_growth**): Governs how sharply peak width decreases with increasing numerator k .
- s_e (**freq_sharpness_exp**): Controls how strongly harmonic order f influences peak sharpness. Larger values make peaks with higher denominators narrower and harder to match.
- d (**decay**): Exponential decay of contribution with increasing k .
- $f_{\max}$ (**max harmonic order**): The highest denominator taken into account.
- w_e (**weight exponent**): Penalizes higher harmonic orders f , reflecting perceptual salience of simple ratios.
- $x_{\max}$ (**max**): Can be set to ignore all k/f above this value when computing the scoring function.

Output scaling: $P(x)$ is normalized such that the peak at $k = 1, f = 1$ equals M (typically $M = 1$).

In summary, the Proximity function provides a continuous, probabilistic mapping from observed IOI ratios to their theoretical likelihood under intentional small-integer timing organization. It allows statistical detection of rhythmic structure even when exact integer relationships are blurred by natural variability, effectively testing whether empirical IOI clusters reproduce small-integer ratios expected from volitional timing behavior.

Application of the Proximity Function

Parameter considerations

The choice of parameters strongly affects the evaluation of the same IOI sequence, and thus must be carefully considered when analysing data using the Proximity function. Depending on the research question, some parameters are more relevant than others:

- When searching only for **isochrony**, any other Gaussian kernels than $f = 1, k = 1$ are not required and get filtered out through $x_{max} = 1$. In this case, parameters such as *decay* d , *sharpness growth* S_g , *weight exponent* w_e , and *frequency sharpness exponent* s_e become irrelevant. The tolerance for timing jitter is mainly determined by *sharpness base* S_b .
- When investigating more complex rhythmic structures, the other parameters become increasingly important, as they shape the sensitivity of the proximity scoring to subtle deviations from small-integer ratios and make assumptions about probability/weighting of ratios.

In this study, we focus on a limited set of five parameter configurations for comparability across datasets. We use a slight *decay* d and *sharpness growth* S_g (both 0.1) and set *weight exponent* w_e to 0 in all configurations. The other parameters are chosen to match specific levels of complexity from isochrony to complex harmonics ($f = 4$) (Fig. 2).

To find other meaningful parameters for a given dataset we included a small “ipywidget”-tool to explore the functions parameters visually in the repository.

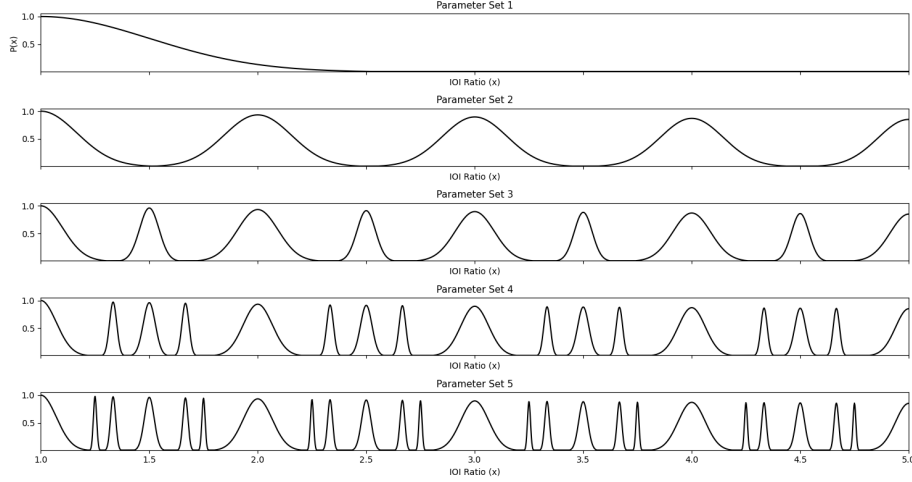


Figure 2: Visualisation of the Proximity function with the 5 parameter configurations used in this paper.

Application to IOI sequences

To determine whether a given IOI sequence (e.g., a sperm whale coda recording) exhibits rhythmic structure, we proceed as follows:

1. Build ratios of all IOI pairs (x_i, x_j) and avoid redundancy through $i > j$.
2. Evaluate each ratio via the Proximity function, producing a set of **unique proximity scores** and their corresponding k/f labels.

Since any sequence produces a list of proximity scores, a **null model** is necessary for comparison. We generate at least 500 null sequences of the same length as the empirical sequence to ensure comparable statistical evaluation.

(Proximity scores in a sequence are typically non-Gaussian, so we compare the *median* of the empirical proximity scores to the *mean of medians* from the null sequences. Metrics of interest include:–)–, nich so sicher was das betrifft den teil muss ich auch aus dem alten code noch in die lib übernehmen

- Difference in proximity score between empirical and null sequences.
- Empirical p -value: fraction of null medians exceeding the empirical median, calculated as $(\# \text{ null medians} > \text{empirical median} + 1) / \# \text{ null sequences}$.
- Z-score: number of standard deviations the empirical median is above the mean of null medians.

was noch fehlt:

Null models?

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ob wir die includen oder nicht

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